

606026



Test Booklet No.

ENGLISH

Test Booklet Code

11

# NEET RANKERS TEST SERIES-2025



Do not open this Test Booklet until you are asked to do so.

## Important Instructions:

- The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with blue/black ball point pen only.
- The test is of 3 hours 20 minutes duration and the Test Booklet contains 200 multiple-choice questions (four options with a single correct answer) from Physics, Chemistry and Biology (Botany and Zoology). 50 questions in each subject are divided into two Sections (A and B) as per details given below:
  - Section A shall consist of 35 (Thirty-five) Questions in each subject (Question Nos - 1 to 35, 51 to 85, 101 to 135 and 151 and 185). All questions are compulsory.
  - Section B shall consist of 15 (Fifteen) questions in each subject (Question Nos - 36 to 50, 86 to 100, 136 to 150, and 186 to 200). In Section B, a candidate needs to attempt any 10 (Ten) questions out of 15 (Fifteen) in each subject.

Candidates are advised to read all 15 questions in each subject of Section B before they start attempting the question paper. In the event of a candidate attempting more than ten questions, the first ten questions answered by the candidate shall be evaluated.
- Each question carries 4 marks. For each correct response, the candidate will get 4 marks. For each incorrect response, one mark will be deducted from the total scores. The maximum marks are 720.
- Use Blue/Black Ball Point Pen only for writing particulars on this page/mark responses on Answer Sheet.
- Rough work is to be done in the space provided for this purpose in the Test Booklet only.
- On completion of the test, the candidate must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
- The CODE for this Booklet is 11. Make sure that the CODE printed on the Original Copy of the Answer Sheet is the same as that on this Test Booklet. In case of discrepancy, the candidate should immediately report the matter to the Invigilator for replacement of both the Test Booklet and the Answer Sheet.
- The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Roll No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
- Use of white fluid for correction is NOT permissible on the Answer Sheet.
- Each candidate must show on-demand his/her Admit Card to the Invigilator.
- No candidate, without special permission of the centre Superintendent or Invigilator, would leave his/her seat.
- The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet twice. Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.
- Use of Electronic/Manual Calculator is prohibited.
- The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
- No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.
- The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.
- Compensatory time of one hour five minutes will be provided for the examination of three hours and 20 minutes duration, whether such candidate (having a physical limitation to write) uses the facility of Scribe or not.

Name of the Candidate (in Capitals): \_\_\_\_\_

Roll Number: In figures \_\_\_\_\_

: In words \_\_\_\_\_

Centre of Examination (in Capitals): \_\_\_\_\_

Candidate's Signature: \_\_\_\_\_ Invigilator's Signature: \_\_\_\_\_

Facsimile Signature stamp of Centre Superintendent

**Physics : Section-A (Q. No. 1 to 35)**

1. An electron and a proton with equal momentum enter perpendicularly into a uniform magnetic field, then
  - (1) the path of proton shall be more curved than that of electron
  - (2) the path of proton shall be less curved than that of electron
  - (3) Both are equally curved
  - (4) path of both will be straight line
2. To reduce the resonant frequency in an L-C-R series circuit with a generator
  - (1) the generator frequency should be reduced
  - (2) another capacitor should be added in parallel to the first
  - (3) the iron core of the inductor should be removed
  - (4) dielectric in the capacitor should be removed
3. The output of a NAND gate is 0,
  - (1) if both inputs are 0
  - (2) if one input is 0 and the other input is 1
  - (3) if both inputs are 1
  - (4) either if both inputs are 1 or if one of the inputs is 1 and the other 0
4. If 'muscle times speed equals power', then what is the ratio of the SI unit and the CGS unit of muscle?
  - (1)  $10^5$
  - (2)  $10^3$
  - (3)  $10^7$
  - (4)  $10^{-5}$
5. A uniform cube is subjected to volume compression. If each side is decreased by 1%, then bulk strain is
 

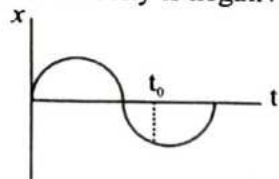
|          |          |
|----------|----------|
| (1) 0.01 | (2) 0.06 |
| (3) 0.02 | (4) 0.03 |
6. The coefficient of friction between the tyres and road is 0.4. The minimum distance covered before attaining a speed of  $8\text{ms}^{-1}$  starting from rest is nearly (take,  $g = 10\text{ms}^{-2}$ )
  - (1) 8 m
  - (2) 4 m
  - (3) 10 m
  - (4) 16 m
7. When a metallic surface is illuminated with radiation of wavelength  $\lambda$ , the stopping potential is  $V$ . If the same surface is illuminated with radiation of wavelength  $2\lambda$ , the stopping potential is  $V/4$ . The threshold wavelength for the metallic surface is
  - (1)  $5\lambda$
  - (2)  $\frac{5}{2}\lambda$
  - (3)  $3\lambda$
  - (4)  $4\lambda$
8. Four capacitors of equal capacitance have an equivalent capacitance  $C_1$  when connected in series and an equivalent capacitance  $C_2$  when connected in parallel. The ratio  $\frac{C_1}{C_2}$  is
  - (1)  $\frac{1}{4}$
  - (2)  $\frac{1}{16}$
  - (3)  $\frac{1}{8}$
  - (4)  $\frac{1}{12}$
9. A stone falls from a balloon that is descending at a uniform velocity of  $12\text{ms}^{-1}$ . The displacement of the stone from the point of release after 10 seconds is: ( $g = 9.8\text{m/s}^2$ )
 

|           |           |
|-----------|-----------|
| (1) 490 m | (2) 510 m |
| (3) 610 m | (4) 750 m |





10. Given below are two statement: one is labelled as Assertion (A) and the other is labelled as Reason (R).  
**Assertion (A):** In the  $x-t$  graph of a particle in SHM acceleration of particle at time  $t_0$  is positive but velocity is negative.



**Reason (R):**  $a \propto -x$  and velocity is slope of  $x-t$  graph.

Where  $a$  is acceleration and  $x$  is displacement.

In the light of the above statements, choose the **correct** answer from the options given below.

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

11. The speed of electromagnetic wave in vacuum
- (1) increases as we move from  $\gamma$ -rays to radio waves
  - (2) decreases as we move from  $\gamma$ -rays to radio waves
  - (3) is same for all of them
  - (4) None of these

12. If  $V$  is the gravitational potential on the surface of the earth, then what is its value at the centre of the earth?

- (1)  $2V$
- (2)  $3V$
- (3)  $\frac{3}{2}V$
- (4)  $\frac{2}{3}V$

13. A spring 40 mm long is stretched by the application of a force. If 10 N force is required to stretch the spring through 1 mm, then the work done in stretching the spring through 40 mm is

- (1) 84 J
- (2) 68 J
- (3) 23 J
- (4) 8 J

14. Calculate the amount of charge flowed in 2 min through a wire of resistance  $10 \Omega$ , when a potential difference of 20 V is applied across its ends.

- (1) 120 C
- (2) 240 C
- (3) 20 C
- (4) 4 C

15. From a disc of radius  $R$  and mass  $M$ , a circular hole of diameter  $R$ , whose rim passes through the centre is cut. What is the moment of inertia of the remaining part of the disc about a perpendicular axis, passing through the centre ?

- (1)  $13 MR^2/32$
- (2)  $11 MR^2/32$
- (3)  $9 MR^2/32$
- (4)  $15 MR^2/32$

16. The ratio of kinetic energy to the total energy of an electron in a Bohr orbit of the hydrogen atom is

- (1) 2 : -1
- (2) 1 : -1
- (3) 1 : 1
- (4) 1 : -2

17. Given below are two statement: one is labelled as Assertion (A) and the other is labelled as Reason (R).

**Assertion (A):** The images formed by total internal reflection are much brighter than those formed by mirrors or lenses.

**Reason (R):** There is no loss of intensity in total internal reflection.

In the light of the above statements, choose the **correct** answer from the options given below.

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

18. In an AC, series  $L$ - $C$ - $R$  circuit,  $R = X_L = X_C$  and applied AC voltage is  $V$ . then match the following two columns and choose the **correct** option from codes given below.

( $V_{xy}$  = potential difference across  $x$  and  $y$ )

**List-I**

**List-II**

- |              |                   |
|--------------|-------------------|
| (A) $V_R$    | (I) zero          |
| (B) $2V_C$   | (II) $V$          |
| (C) $V_{RL}$ | (III) $\sqrt{2}V$ |
| (D) $V_{CL}$ | (IV) $2V$         |

Choose the **correct** answer from the options given below.

- (1) A-II, B-IV, C-III, D-I  
 (2) A-I, B-IV, C-II, D-III  
 (3) A-II, B-I, C-III, D-IV  
 (4) A-II, B-III, C-IV, D-I

19. Which of the following has negative temperature coefficient of resistance?

- (1) Copper  
 (2) Aluminium  
 (3) Iron  
 (4) Germanium

20. A particle of mass 2 kg is moving along a circular path of radius 1 m. If its angular speed is  $2\pi$  rad  $s^{-1}$ , the centripetal force on it is

- (1)  $4\pi$  N                      (2)  $8\pi$  N  
 (3)  $4\pi^2$  N                    (4)  $8\pi^2$  N

21. When a uranium isotope  ${}_{92}^{235}U$  is bombarded with a neutron, it generates  ${}_{36}^{89}Kr$ , three neutrons and

- (1)  ${}_{40}^{91}Zr$                       (2)  ${}_{36}^{101}Kr$   
 (3)  ${}_{36}^{103}Kr$                     (4)  ${}_{56}^{144}Ba$

22. The ratio of escape velocity at earth ( $v_e$ ) to the escape velocity at a planet ( $v_p$ ) whose radius and mean density are twice as of earth is

- (1)  $1:2\sqrt{2}$                     (2)  $1:4$   
 (3)  $1:\sqrt{2}$                     (4)  $1:2$

23. When a glass rod is rubbed with silk, it  
 (1) gains electrons from silk  
 (2) gives electrons to silk  
 (3) gains protons from silk  
 (4) gives protons to silk

24. A current carrying infinitely long wire is kept along the diameter of a circular wire loop, without touching it, the correct statement(s) is(are);

- A. The emf induced in the loop is zero if the current is constant.  
 B. The emf induced in the loop is finite if the current is constant.  
 C. The emf induced in the loop is zero if the current decreases at a steady rate.

Choose the most appropriate answer from the options given below.

- (1) A only  
 (2) B only  
 (3) A and C  
 (4) A, B and C

25. The momentum  $p$  (in  $kg\ ms^{-1}$ ) of a particle is varying with time  $t$  (in second) as  $p = 2 + 3t^2$  The force acting on the particle at  $t = 3s$  will be

- (1) 18 N  
 (2) 54 N  
 (3) 9 N  
 (4) 15 N

26. Two slits are separated by a distance of 0.5 mm and illuminated with light of wave length  $\lambda = 6000\ \text{\AA}$ . If the screen is placed 2.5 m from the slits. The distance of the third bright image from the centre will be

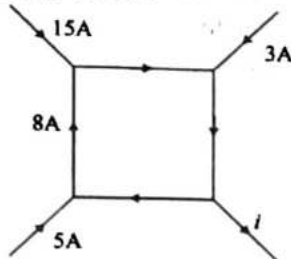
- (1) 1.5 mm                      (2) 3 mm  
 (3) 6 mm                        (4) 9 mm

27. Moment of a force of magnitude 20 N acting along positive  $x$ -direction at point  $(3m, 0, 0)$  about the point  $(0, 2m, 0)$  in  $N\text{-m}$  is

- (1) 20                              (2) 60  
 (3) 40                              (4) 30



28. The figure shows a network of currents. The current  $i$  will be

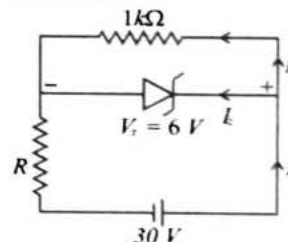


- (1) 3 A (2) 13 A  
(3) 23 A (4) -3 A
29. The fundamental frequency of a closed organ pipe of length 20 cm is equal to the second overtone of an organ pipe open at both the ends. The length of organ pipe open at both the ends is  
(1) 80 cm (2) 100 cm  
(3) 120 cm (4) 140 cm
30. If the frequency of light in a photoelectric experiment is doubled, then the stopping potential will  
(1) be doubled  
(2) be halved  
(3) become more than double  
(4) become less than double
31. The dimensional formula for electric flux is  
(1)  $[ML^3I^{-1}T^{-3}]$  (2)  $[ML^2I^{-1}T^{-2}]$   
(3)  $[ML^3I^{-1}T^{-3}]$  (4)  $[ML^{-3}I^{-1}T^{-3}]$
32. A spherical conductor of radius 2 m is charged to a potential of 120 V. It is now placed inside another hollow spherical conductor of radius 6 m. Calculate the potential to which the bigger sphere would be raised.  
(1) 20 V (2) 60 V  
(3) 80 V (4) 40 V
33. A cyclist bends while taking turn in order to  
(1) reduce friction  
(2) provide required centripetal force  
(3) reduce apparent weight  
(4) reduce speed

34. The kinetic energy acquired by a body of mass  $m$  in travelling a certain distance starting from rest, under a constant force is  
(1) directly proportional to  $m$   
(2) directly proportional to  $\sqrt{m}$   
(3) inversely proportional to  $\sqrt{m}$   
(4) independent of  $m$
35. A long solenoid has 500 turns. When a current of 2A is passed through it, then the resulting magnetic flux linked with each turn of the solenoid is  $4 \times 10^{-3}$  Wb. The self-inductance of the solenoid is  
(1) 1.0 H (2) 4.0 H  
(3) 2.5 H (4) 2.0 H

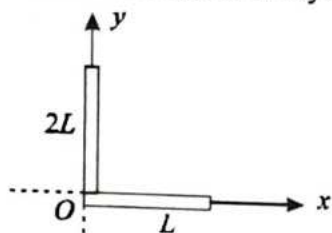
**Physics : Section-B (Q. No. 36 to 50)**

36. The refractive indices of glass and water with respect to air are  $3/2$  and  $4/3$ , respectively. The refractive index of glass with respect to water will be  
(1)  $8/9$  (2)  $9/8$   
(3)  $7/6$  (4) 2
37. A cube of side  $a$  is placed in a uniform electric field  $E = E_0\hat{i} + E_0\hat{j} + E_0\hat{k}$ . Total electric flux passing through the cube would be  
(1)  $E_0a^2$  (2)  $2E_0a^2$   
(3)  $6E_0a^2$  (4) zero
38. If voltage across a zener diode is 6V, then find out the value of maximum resistance in this condition.



- (1) 2 kΩ (2) 3 kΩ  
(3) 5 kΩ (4) 4 kΩ

39. Figure shows a composite system of two uniform rods of lengths as indicated. Then the coordinates of the centre of mass of the system of rods are



- (1)  $\left(\frac{L}{2}, \frac{2L}{3}\right)$  (2)  $\left(\frac{L}{4}, \frac{2L}{3}\right)$   
 (3)  $\left(\frac{L}{6}, \frac{2L}{3}\right)$  (4)  $\left(\frac{L}{6}, \frac{L}{3}\right)$
40. Steam at  $100^\circ\text{C}$  is passed into 20 g of water at  $10^\circ\text{C}$ . When water acquires a temperature of  $80^\circ\text{C}$ , the mass of water present will be [Take, specific heat capacity of water  $1 \text{ cal g}^{-1} ^\circ\text{C}^{-1}$  and latent heat of steam  $= 540 \text{ cal g}^{-1}$ ]
- (1) 24 g  
 (2) 31.5 g  
 (3) 42.5 g  
 (4) 22.5 g

41. Given below are two statements.  
**Statement-I:** Susceptibilities of paramagnetic and ferromagnetic substances increase with decrease in temperature.  
**Statement-II:** Diamagnetism is a result of orbital motions of electrons developing magnetic moments opposite to the applied magnetic field.  
 In the light of the above statements, choose the most appropriate answer from the options given below.
- (1) Statement-I is correct but Statement-II is incorrect.  
 (2) Statement-I is incorrect but Statement-II is correct.  
 (3) Both Statement-I and Statement-II are correct.  
 (4) Both Statement-I and Statement-II are incorrect.

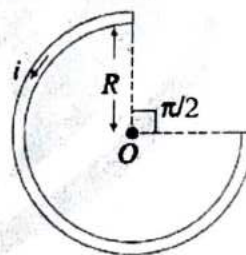
42.  $P$ - $V$  diagram of a diatomic gas is a straight line passing through origin. The molar heat capacity of the gas in the process will be

- (1)  $4R$   
 (2)  $2.5R$   
 (3)  $3R$   
 (4)  $\frac{4R}{3}$

43. For a parallel beam of monochromatic light of wavelength  $\lambda$  diffraction is produced by a single slit whose width  $a$  is of the order of the wavelength of the light. If  $D$  is the distance of the screen from the slit, the width of the central maximum will be

- (1)  $\frac{2D\lambda}{a}$  (2)  $\frac{D\lambda}{a}$   
 (3)  $\frac{Da}{\lambda}$  (4)  $\frac{2Da}{\lambda}$

44. A current  $i$  ampere flows in a circular arc of radius  $R$ , which subtends an angle  $3\pi/2$  radian at its centre. The magnetic induction  $B$  at the centre is



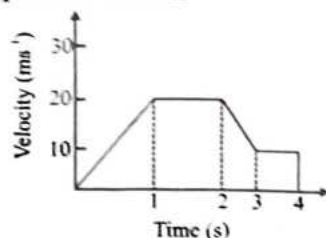
- (1)  $\frac{\mu_0 i}{R}$  (2)  $\frac{\mu_0 i}{2R}$   
 (3)  $\frac{2\mu_0 i}{R}$  (4)  $\frac{3\mu_0 i}{8R}$

45. Two point charges  $+8q$  and  $-2q$  are located at  $x = 0$  and  $x = L$  respectively. The location of a point on the  $X$ -axis at which the net electric field due to these two point charges is zero, is

- (1)  $8L$  (2)  $4L$   
 (3)  $2L$  (4)  $L/4$



46. The variation of velocity of a particle with time moving along a straight line is illustrated in the adjoining figure. The distance travelled by the particle in 4s is



- (1) 60 m (2) 55 m  
(3) 25 m (4) 30 m
47. A gas is found to obey the law  $p^2V = \text{constant}$ . The initial temperature and volume are  $T_0$  and  $V_0$ . If the gas expands to a volume  $3V_0$  its final temperature becomes

- (1)  $\frac{T_0}{3}$  (2)  $\frac{T_0}{\sqrt{3}}$   
(3)  $3T_0$  (4)  $\sqrt{3}T_0$

48. The horizontal range of a projectile is  $4\sqrt{3}$  times of its maximum height. The angle of projection will be

- (1)  $60^\circ$  (2)  $37^\circ$   
(3)  $30^\circ$  (4)  $45^\circ$

49. The electric and the magnetic field associated with an E.M. wave, propagating along the  $+z$ -axis, can be represented by;

- (1)  $[\vec{E} = E_0\hat{i}, \vec{B} = B_0\hat{j}]$   
(2)  $[\vec{E} = E_0\hat{k}, \vec{B} = B_0\hat{i}]$   
(3)  $[\vec{E} = E_0\hat{j}, \vec{B} = B_0\hat{i}]$   
(4)  $[\vec{E} = E_0\hat{j}, \vec{B} = B_0\hat{k}]$

50. A rod of weight  $w$  is supported by two parallel knife edges  $A$  and  $B$  and is in equilibrium in a horizontal position. The knives are at a distance  $d$  from each other. The centre of mass of the rod is at distance  $x$  from  $A$ . The normal reaction on  $A$  is

- (1)  $\frac{wx}{d}$  (2)  $\frac{wd}{x}$   
(3)  $\frac{w(d-x)}{x}$  (4)  $\frac{w(d-x)}{d}$

### Chemistry : Section-A (Q. No. 51 to 85)

51. The correct order of decreasing polarizability of ion are;

- (1)  $\text{Cl}^- > \text{Br}^- > \text{I}^- > \text{F}^-$   
(2)  $\text{F}^- > \text{I}^- > \text{Br}^- > \text{Cl}^-$   
(3)  $\text{I}^- > \text{Br}^- > \text{Cl}^- > \text{F}^-$   
(4)  $\text{F}^- > \text{Cl}^- > \text{Br}^- > \text{I}^-$

52. The electron affinity value are negative for:

- A.  $\text{Be} \rightarrow \text{Be}^-$  B.  $\text{N} \rightarrow \text{N}^-$   
C.  $\text{O} \rightarrow \text{O}^{2-}$  D.  $\text{Na} \rightarrow \text{Na}^-$   
E.  $\text{Al} \rightarrow \text{Al}^-$

Choose the most appropriate answer from the options given below:

- (1) (D) and (E) only  
(2) (A), (B), (D) and (E) only  
(3) (A) and (D) only  
(4) (A), (B) and (C) only

53. The conjugate base of  $\text{H}_2\text{PO}_4^-$  is:

- (1)  $\text{H}_3\text{PO}_4$  (2)  $\text{H}_2\text{PO}_4^-$   
(3)  $\text{HPO}_4^{2-}$  (4)  $\text{PO}_4^{3-}$

54. Oxidation state of oxygen in  $\text{H}_2\text{O}_2$  is:

- (1)  $-1$  (2)  $+2$   
(3)  $+\left(\frac{1}{2}\right)$  (4)  $-2$

55. Daniell cell is represented as:

- (1)  $\text{Zn(s)} | \text{Zn}^{2+}(\text{aq}) || \text{Cu}^{2+}(\text{aq}) | \text{Cu(s)}$   
(2)  $\text{Cu(s)} | \text{Cu}^{2+}(\text{aq}) || \text{Zn}^{2+}(\text{aq}) | \text{Zn(s)}$   
(3)  $\text{Zn(s)} | \text{Zn}^{2+}(\text{aq}) || \text{Zn}^{2+}(\text{aq}) | \text{Zn(s)}$   
(4)  $\text{Cu(s)} | \text{Cu}^{2+}(\text{aq}) || \text{Cu}^{2+}(\text{aq}) | \text{Cu(s)}$

56. For the reaction,  $N_2 + 3H_2 \rightarrow 2NH_3$ , rate is expressed as:

$$\begin{aligned} (1) \quad & \frac{1}{3} \frac{d[N_2]}{dt} = -\frac{1}{2} \frac{d[H_2]}{dt} = 3 \frac{d[NH_3]}{dt} \\ (2) \quad & -\frac{d[N_2]}{dt} = -\frac{1}{3} \frac{d[H_2]}{dt} = \frac{1}{2} \frac{d[NH_3]}{dt} \\ (3) \quad & -\frac{d[N_2]}{dt} = -3 \frac{d[H_2]}{dt} = 2 \frac{d[NH_3]}{dt} \\ (4) \quad & -\frac{d[N_2]}{dt} = -\frac{d[H_2]}{dt} = \frac{d[NH_3]}{dt} \end{aligned}$$

57. Given below are two statements:

**Statement-I:** Deficiency of vitamin  $B_1$  causes the disease Beri-beri.

**Statement-II:** Vitamin D is a water soluble vitamin.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement-I is correct but Statement-II is incorrect.
- (2) Statement-I is incorrect but Statement-II is correct.
- (3) Both Statement-I and Statement-II are correct.
- (4) Both Statement-I and Statement-II are incorrect.

58. What is the maximum amount of nitrogen dioxide that can be produced by mixing 4.2 g of  $NO(g)$  and 3.2 g of  $O_2(g)$ ?

- (1) 4.60 g
- (2) 2.30 g
- (3) 3.22 g
- (4) 6.44 g

59. Given below are two statements:

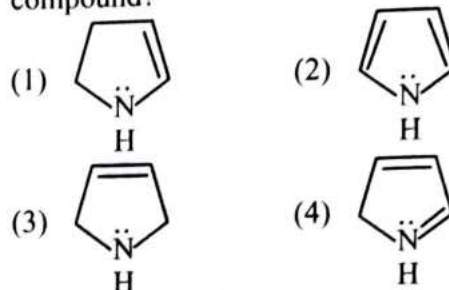
**Statement-I:** First ionization energy of P is greater than S.

**Statement-II:** First ionization energy of Na is lesser than Mg.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement-I is correct but Statement-II is incorrect.
- (2) Statement-I is incorrect but Statement-II is correct.
- (3) Both Statement-I and Statement-II are correct.
- (4) Both Statement-I and Statement-II are incorrect.

60. Which of the following is an aromatic compound?



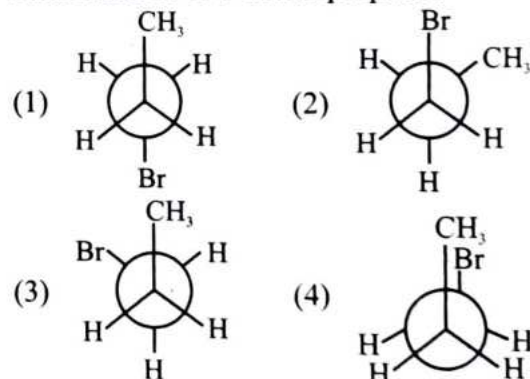
61. What will be the molality of a solution having 18 g glucose (mol. wt. =  $180 \text{ g mol}^{-1}$ ) is dissolved in 500 g of water?

- (1) 1 m
- (2) 0.5 m
- (3) 0.2 m
- (4) 2 m

62. The standard reduction potential values of three metallic cations X, Y and Z are 0.52, -3.03 and -1.18 V respectively. The order of reducing power of the corresponding metal is:

- (1)  $Y > Z > X$
- (2)  $X > Y > Z$
- (3)  $Z > Y > X$
- (4)  $Z > X > Y$

63. Which of the following is the most stable conformation of 1-bromopropane?







64. Which of the following group is known as chalcogens?  
 (1) B, Al, Ga, In (2) N, P, As, Sb  
 (3) F, Cl, Br, I (4) O, S, Se, Te
65. EDTA is an example of;  
 (1) monodentate ligand  
 (2) bidentate ligand  
 (3) tridentate ligand  
 (4) hexadentate ligand
66. Which of the following is/are V-shaped?  
 (I)  $\text{SO}_2$   
 (II)  $\text{I}_3^-$   
 (III)  $\text{NO}_2^+$   
 (IV)  $\text{I}_3^+$   
 (1) III and IV (2) II and IV  
 (3) I and IV (4) I and II
67. Given below are two statements.  
**Statement-I:** The number of unpaired electrons in  $\text{Fe}^{3+}$  are 5.  
**Statement-II:** The number of unpaired electrons in  $\text{Cr}^{2+}$  is 4.  
 In the light of the above statements, choose the most appropriate answer from the options given below:  
 (1) Statement-I is correct but Statement-II is incorrect.  
 (2) Statement-I is incorrect but Statement-II is correct.  
 (3) Both Statement-I and Statement-II are correct.  
 (4) Both Statement-I and Statement-II are incorrect.
68. The pH of an aqueous solution of 1.0 M ammonium formate assuming complete dissociation is: ( $\text{pK}_a$  of formic acid = 3.8 and  $\text{pK}_b$  of ammonia = 4.8)  
 (1) 7.0 (2) 7.5  
 (3) 6.5 (4) 4.3

69. It takes 10 min for the decomposition of 50%  $\text{H}_2\text{O}_2$ . If the reaction is of first order, the rate constant will be:  
 (1) 0.693  
 (2) 0.00693  
 (3) 0.0693  
 (4) 6.93

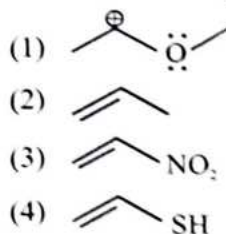
70. Match List-I with List-II.

| List-I<br>(Compound)        | List-II<br>(Nature) |
|-----------------------------|---------------------|
| (A) $\text{Na}_2\text{O}$   | (I) Neutral         |
| (B) $\text{Al}_2\text{O}_3$ | (II) Basic          |
| (C) $\text{N}_2\text{O}$    | (III) Acidic        |
| (D) $\text{Cl}_2\text{O}_7$ | (IV) Amphoteric     |

Choose the **correct** answer from the options given below:

- (1) A-IV, B-II, C-III, D-I  
 (2) A-III, B-II, C-IV, D-I  
 (3) A-II, B-I, C-III, D-IV  
 (4) A-II, B-IV, C-I, D-III
71. The IUPAC name of  $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$  is:  
 (1) Cobalthexaminechloride  
 (2) Hexaamminecobalt (III) chloride  
 (3) Hexaamminocobalto chloride  
 (4) Hexaaminocobaltate chloride
72. Entropy is a:  
 (1) state function and extensive property.  
 (2) state function and intensive property.  
 (3) path function and extensive property.  
 (4) path function and intensive property.

73. Resonance is **not** possible in:



74. In a chemical reaction:  $A + B \rightarrow AB$ ,  
If B is acting as a limiting reagent, then choose the correct option.

| A             | B         |
|---------------|-----------|
| (1) 50 atoms  | 100 atoms |
| (2) 100 atoms | 200 atoms |
| (3) 50 atoms  | 30 atoms  |
| (4) 50 atoms  | 200 atoms |

75. Given below are two statement: one is labelled as Assertion (A) and the other is labelled as Reason (R):

**Assertion (A):** Hydrolysis of ethyl acetate is a pseudo first order reaction.

**Reason (R):** Water is present in large excess and therefore its concentration remain constant during the course of reaction.

In the light of the above statements, choose the correct answer from the options given below:

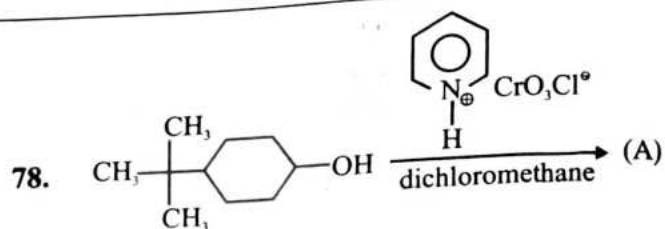
- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

76. What would be the product when isopropyl chloride reacts with KCN?

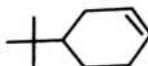
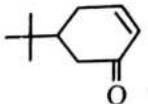
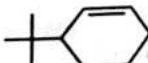
- (1) Propene
- (2) 2-Methylpropanenitrile
- (3) Propyne
- (4) Propanenitrile

77. The volume of a system becomes twice its original volume on the absorption of 300 cal of heat. The work done on the surrounding was found to be 200 cal. What is the change in internal energy ( $\Delta U$ ) of the system?

- (1) 500 cal
- (2) 300 cal
- (3) 100 cal
- (4) -500 cal

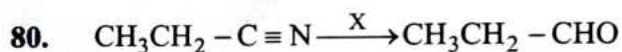


What is the major product (A) in the above reaction?

- (1) 
- (2) 
- (3) 
- (4) 

79. Stark effect refers to the:

- (1) Splitting up of the spectral lines in an emission spectrum in the presence of external electrostatic field.
- (2) Random scattering of light by colloidal Particles.
- (3) Splitting up of the spectral lines in an emission spectrum in a magnetic field.
- (4) Emission of electrons from metals when light falls upon them.



The reagent 'X' used here is (are):

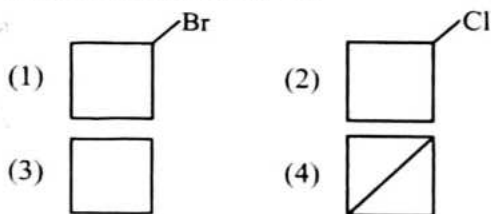
- (1)  $\text{SnCl}_2/\text{HCl}$ ,  $\text{H}_2\text{O}/\text{boil}$
- (2)  $\text{H}_2/\text{Pd}-\text{BaSO}_4$
- (3)  $\text{LiAlH}_4/\text{ether}$
- (4)  $\text{NaBH}_4/\text{ether}$ ,  $\text{H}_3\text{O}^+$

81. What current strength in ampere will be required to liberate 10 g of chlorine from sodium chloride solution in one hour?

- (1) 7.55 ampere
- (2) 7.15 ampere
- (3) 7.50 ampere
- (4) 7.45 ampere



82. What will be the product form when 1-bromo-3-chlorocyclobutane reacts with two equivalents of metallic sodium in ether?



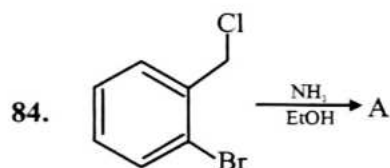
83. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R):

**Assertion (A):** Aromatic carboxylic acids do not undergo Friedel-Craft reaction.

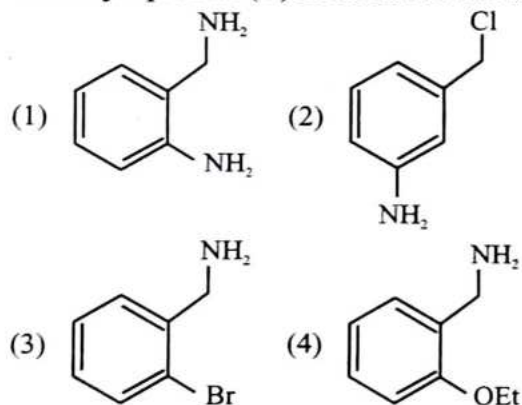
**Reason (R):** Carboxyl group is meta-directing group.

In the light of the above statements, choose the correct answer from the options given below:

- (1) A is true but R is false.  
 (2) A is false but R is true.  
 (3) Both A and R are true and R is the correct explanation of A.  
 (4) Both A and R are true but R is NOT the correct explanation of A.



The major product (A) in the above reaction is:



85. If wavelength is equal to the distance travelled by the electron in one second, then:

(1)  $\lambda = \frac{h}{p}$  (2)  $\lambda = \frac{h}{m}$   
 (3)  $\lambda = \sqrt{\frac{h}{p}}$  (4)  $\lambda = \sqrt{\frac{h}{m}}$

**Chemistry : Section-B (Q. No. 86 to 100)**

86. Glucose reduces:

- (1) tollen's reagent  
 (2) fehling's solution  
 (3) benedict's solution  
 (4) all of these

87. For a process to be spontaneous at constant T and P;

- (1)  $(\Delta G)_{\text{system}}$  must be negative  
 (2)  $(\Delta G)_{\text{system}}$  must be positive  
 (3)  $(\Delta S)_{\text{system}}$  must be zero  
 (4)  $(\Delta S)_{\text{system}}$  must be negative

88. Which of the following mixture does **not** show positive deviation from the Raoult's law?

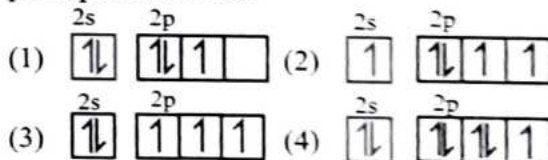
- (1) Benzene + acetone  
 (2) Acetone + ethanol  
 (3) Acetone + chloroform  
 (4) Water + benzene

89. Which of the following statement (s) are correct?

- A. Nitrogen atom of amines is trivalent and carries an unshared pair of electrons.  
 B. Nitrogen orbitals in amines are  $sp^3$  hybridised.  
 C. The geometry of amines is pyramidal.  
 D. The angle C-N-E, (where E is C or H) is more than  $109.5^\circ$ .

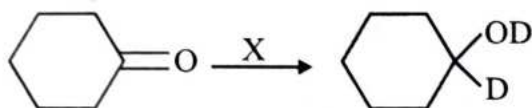
- (1) A, B and C (2) A and D  
 (3) A, C and D (4) B, C and D

90. The orbital diagram in which the Aufbau principle is violated:



91. Which of the following statement (s) are correct?
- A. n-pentane has higher boiling point than neopentane.
- B. The rate of reaction of alkanes with halogens is  $F_2 > Cl_2 > Br_2 > I_2$ .
- C. Rate of replacement of hydrogens of alkanes is :  $3^\circ > 2^\circ > 1^\circ$ .
- D. Boiling point increases with increase in branching.
- (1) A, B and C      (2) A and D  
(3) A, C and D      (4) B, C and D

92. In the given reaction:



[X] will be:

- (1)  $LiAlD_4/D_2O$       (2)  $NaBD_4/H_2O$   
(3)  $LiAlD_4/HOH$       (4)  $NaBH_4/D_2O$

93. Which of the following pairs of compounds are **not** position isomers?

- (1) and
- (2) and
- (3) and
- (4) and

94. Match List-I with List-II.

| List-I<br>(Structure) | List-II<br>(Common name) |
|-----------------------|--------------------------|
| (A)                   | (I) Neopentyl alcohol    |
| (B)                   | (II) Neopentyl thiol     |

- (C) (III) Formaldehyde
- (D) (IV) Isopentane

Choose the **correct** answer from the options given below:

- (1) A-III, B-I, C-II, D-IV  
(2) A-IV, B-I, C-III, D-II  
(3) A-II, B-I, C-III, D-IV  
(4) A-II, B-III, C-IV, D-I

95. Given below are two statement: one is labelled as Assertion (A) and the other is labelled as Reason (R):

**Assertion (A):** Ammonolysis of alkyl halides is not suitable method for the preparation of pure primary amines.

**Reason (R):** Ammonolysis of alkyl halides yields mainly secondary amines.

In the light of the above statements, choose the **correct** answer from the options given below:

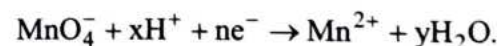
- (1) A is true but R is false.  
(2) A is false but R is true.  
(3) Both A and R are true and R is the correct explanation of A.  
(4) Both A and R are true but R is NOT the correct explanation of A.

96.  $\xrightarrow[H^+]{LiAlH_4}$  product;

The type of above reaction is;

- (1) Electrophilic addition reaction  
(2) Nucleophilic addition reaction  
(3) Nucleophilic substitution reaction  
(4) Electrophilic substitution reaction

97. In the reaction:



What is the value of n?

- (1) 5      (2) 8  
(3) 6      (4) 3



98. The colour of ppt. formed in sulphate ion test is:

- (1) yellow (2) white  
(3) black (4) red

99. What is the molarity of NaOH solution if 250 mL of it contains 1 mg of NaOH?

(Given molar mass of NaOH = 40 g/mol)

- (1)  $10^{-1}$  M (2)  $10^{-2}$  M  
(3)  $10^{-4}$  M (4)  $10^{-3}$  M

100. Wilkinson's catalyst is:

- (1)  $\text{TiCl}_4$  and  $\text{Al}(\text{C}_2\text{H}_5)_3$   
(2)  $[\text{Rh}(\text{Cl})(\text{PPh}_3)_3]$   
(3)  $\text{TiCl}_4$  and  $(\text{C}_6\text{H}_5)_3\text{P}$   
(4)  $\text{Al}(\text{C}_2\text{H}_5)_3$ ,  $\text{TiCl}_4$  and  $(\text{C}_6\text{H}_5)_3\text{P}$

**Botany : Section-A (Q. No. 101 to 135)**

101. Isogamous type of sexual reproduction takes place in;

- (1) *Volvox*. (2) *Fucus*.  
(3) *Eudorina*. (4) *Spirogyra*.

102. During light reaction in photosynthesis, the following are formed;

- (1) ATP and sugar.  
(2) Hydrogen,  $\text{O}_2$  and sugar.  
(3) ATP, NADPH and  $\text{O}_2$ .  
(4) ATP, hydrogen and  $\text{O}_2$  donor.

103. Exploration of molecular, genetic and species level diversity for novel products of economic importance is known as;

- (1) narrowly utilitarian.  
(2) broadly utilitarian.  
(3) endemism.  
(4) bioprospecting.

104. Select **correct** option w.r.t the cells which lack nucleus at maturity in humans and plants, respectively.

- (1) Erythrocytes and sieve tube cells  
(2) Cheek cells and sieve tube cells  
(3) Erythrocytes and mesophyll cells  
(4) Cheek cells and mesophyll cells

105. Identify the placentation from the given diagrams and give one example of each.



A

B

C

A

B

C

- |                      |                           |                               |
|----------------------|---------------------------|-------------------------------|
| (1) Marginal, pea    | Parietal, <i>Argemone</i> | Basal, wheat                  |
| (2) Axile, lemon     | Basal, marigold           | Free central, <i>Dianthus</i> |
| (3) Basal, sunflower | Parietal, marigold        | Free central, pea             |
| (4) Parietal, radish | Basal, maize              | Marginal, tomato              |

106. Gene regulation in eukaryotes exerted at:

- (1) transcriptional level.  
(2) processing level.  
(3) translational level.  
(4) All of these

107. Given below are two statements:

**Statement-I:** In some flowers, the calyx and corolla are not distinct and are termed as perianth.

**Statement-II:** In compound leaves, incisions of the lamina reach up to the midrib, breaking it into a number of leaflets.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement-I is correct but Statement-II is incorrect.  
(2) Statement-I is incorrect but Statement-II is correct.  
(3) Both Statement-I and Statement-II are correct.  
(4) Both Statement-I and Statement-II are incorrect.

108. Sequence the following events according to their occurrence in the light reaction of photosynthesis:

- ATP synthesis
- Water splitting
- Electron transfer to  $\text{NADP}^+$
- Excitation of electrons by light
- Release of oxygen

Choose the correct sequence from the options given below:

- (d)-(b)-(a)-(c)-(e)
- (d)-(b)-(e)-(a)-(c)
- (b)-(d)-(e)-(a)-(c)
- (b)-(d)-(a)-(c)-(e)

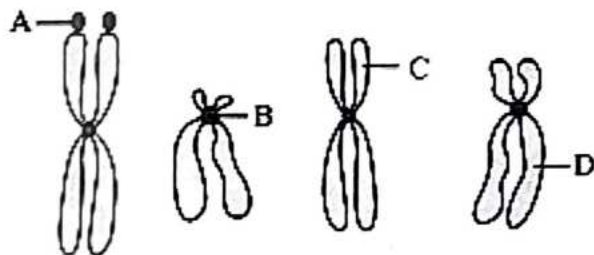
109. Given below are some statements (A-D).

- RNAi takes place in all prokaryotic organisms as a method of cellular defense.
- The proteins encoded by the genes *cryIAc* and *cryIIAb* control the cotton bollworms.
- The source of complementary RNA in RNAi could be from an infection by viruses having RNA genomes.
- Using retrovirus, the nematode-specific genes were introduced into the tobacco plant.

Choose the **incorrect** statements from the options given below:

- A and C only
- A and B only
- C and D only
- A and D only

110. Find out the **correct** option on the basis of following diagrams.



- A-Satellite, B-Secondary constriction, C-Short arm, D- Long arm
- A-Satellite, B-Centromere, C-Short arm, D- Long arm
- A-Secondary constriction, B - Satellite, C-Long arm, D - Short arm
- A-Centromere, B-Secondary constriction, C-Short arm, D-Long arm

111. What type of pollination takes place in *Vallisneria*?

- Male flowers are carried passively by water currents to female flowers at surface of water.
- Pollination occurs in submerged condition in water.
- Flowers emerge above surface of water and pollination occurs by insects.
- Flowers emerge above water surface and pollen is carried by wind.

112. In  $\text{C}_3$  plant, when  $\text{O}_2$  concentration is more, the  $\text{O}_2$  binds to RuBisCO and RuBP gets changed to;

- 2 molecules of PGA.
- 2 molecules of phosphoglycerate.
- 2 molecules of phosphoglycolate.
- one molecule each of phosphoglycerate and phosphoglycolate.

113. Given below are two statements:

**Statement-I:** Double stranded polynucleotide chains of DNA have anti-parallel polarity.

**Statement-II:** In RNA, every nucleotide residue has an additional  $-\text{OH}$  group present at  $2'$  - position in the ribose.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

- Statement-I is correct but Statement-II is incorrect.
- Statement-I is incorrect but Statement-II is correct.
- Both Statement-I and Statement-II are correct.
- Both Statement-I and Statement-II are incorrect.

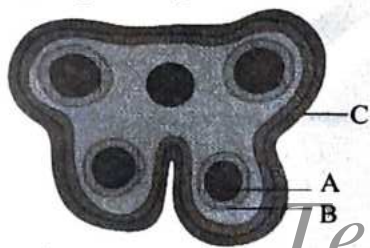




114. Telophase is characterized by all the following **except**;

- (1) chromosomes cluster at opposite spindle poles.
- (2) identity of chromosomes is lost as discrete elements.
- (3) nuclear envelope develops around the chromosome clusters at each pole.
- (4) nucleolus, golgi complex and ER disappear.

115. Identify the labellings from A to C in the given diagram, respectively.



- (1) A-Sporogenous tissue, B-Tapetum, C-Epidermis
- (2) A-Sporogenous tissue, B-Epidermis, C-Tapetum
- (3) A-Epidermis, B-Sporogenous tissue, C-Tapetum
- (4) A-Tapetum, B-Sporogenous tissue, C-Epidermis

116. Which bacterium helps in the production of 'Swiss cheese'?

- (1) *Propionibacterium sharmanii*
- (2) *Trichoderma polysporum*
- (3) *Saccharomyces cerevisiae*
- (4) *Aspergillus niger*

117. In Mendel's experiments with garden pea, round seed shape (RR) was dominant over wrinkled seeds (rr), yellow seed colour (YY) was dominant over green seed colour (yy).

What is/are the expected phenotypes in the  $F_1$  generation of the cross  $RRYY \times rryy$ ?

- (1) All wrinkled green coloured seeds.
- (2) All wrinkled yellow coloured seeds.
- (3) All round yellow coloured seeds.
- (4) 50% round yellow coloured seeds and 50% wrinkled green colour seeds.

118. A. The productivity of the oceans is greater than land.  
 B. By the process of leaching, water-soluble inorganic nutrients go down into the soil horizon and get precipitated as unavailable salts.  
 C. Catabolism is the process in which bacterial and fungal enzymes degrade detritus into simpler inorganic substances.  
 D. The humus is further degraded by some microbes and release of inorganic nutrients occur by the process known as mineralisation.

Choose the **correct** code;

- (1) A and B only
- (2) B and C only
- (3) B, C and D only
- (4) A, B, C and D

119. Given below are two statements:

**Statement-I:** Endarch and exarch are the terms often used for describing the position of secondary xylem in the plant body.

**Statement-II:** Exarch condition is the common feature of the root system.

In the light of the above statement, choose the *most appropriate* answer from the option given below:

- (1) Statement-I is correct but Statement-II is incorrect.
- (2) Statement-I is incorrect but Statement-II is correct.
- (3) Both Statement-I and Statement-II are correct.
- (4) Both Statement-I and Statement-II are incorrect.

120. How many statements are incorrect?

- An orchid growing on a mango branch represents amensalism.
- Orchids are benefitted but mango trees are harmed.
- Orchid is an epiphyte.
- Mango trees that have orchids growing on them to get support from the tree but not food.

Select the correct option:

- One
- Two
- Three
- Four

121. Given below are two statements.

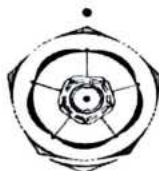
**Statement-I:** Auxin and cytokinins are antagonistic to each other in case of apical dominance.

**Statement-II:** Auxin is not a plant growth regulator.

In the light of the above statements, choose the *most appropriate* answer from the options given below.

- Statement-I is correct but Statement-II is incorrect.
- Statement-I is incorrect but Statement-II is correct.
- Both Statement-I and Statement-II are correct.
- Both Statement-I and Statement-II are incorrect.

122. Identify the correct statement about the diagram.



- The flower of this family consists of a capitulum or head.
- The flower of this family consists of a free anther but fused filaments.
- The flower of this family lacks pappus.
- The flower of this family consists gynoecium of bilocular basal placentation.

123. Match List-I with List-II:

| List-I                               | List-II                                |
|--------------------------------------|----------------------------------------|
| (A) <i>Agrobacterium tumefaciens</i> | (I) Fungus cell wall digesting enzyme  |
| (B) Cellulase                        | (II) Cloning vector                    |
| (C) Ribonuclease                     | (III) Plant cell wall digesting enzyme |
| (D) Chitinase                        | (IV) RNA digesting enzyme              |

Choose the **correct** answer from the options given below:

- A- I, B-IV, C-II, D-III
- A-II, B-III, C-IV, D-I
- A-IV, B-II, C-I, D-III
- A-IV, B-I, C-III, D-II

124. What is **true** about BOD?

- BOD is directly proportional to the organic matter.
- BOD is inversely proportional to organic matter.
- Greater the BOD of waste water, less it's align potential.
- All of these.

125. How many true-breeding pea plant varieties did Mendel select as pairs, which were similar except in one character with contrasting traits?

- 18 only
- 19 only
- 14 only
- 13 only

126. Match the List-I with List-II.

| List-I                | List-II                                                                 |
|-----------------------|-------------------------------------------------------------------------|
| A. Gregor Mendel      | I. Mapped gene positions on chromosomes using recombination frequencies |
| B. Thomas Hunt Morgan | II. Noted the behaviour of chromosomes was                              |



- parallel to the behaviour of genes
- C. Walter Sutton and Theodore Boveri III. Proposed the laws of inheritance in living organisms
- D. Alfred Sturtevant IV. Discovered the basis for the variation that sexual reproduction produced

Choose the **correct** answer from the options given below:

- (1) A-IV, B-III, C-I, D-II  
 (2) A-IV, B-I, C-II, D-III  
 (3) A-III, B-IV, C-II, D-I  
 (4) A-II, B-IV, C-III, D-I

127. What factors contribute to the variability of primary productivity among different ecosystems?

- (a) Type of plant species present  
 (b) Climatic conditions  
 (c) Photosynthetic capacity of plants  
 (d) Animal species present  
 (e) Availability of nutrients

Correct Answer Options:

- (1) (a), (b), (c) and (e) (2) (b), (c) and (d)  
 (3) (a), (c), (d) and (e) (4) (c), (d) and (e)

128. Which of the following is **correct** about venation?

- A. Parallel venation is generally found in monocot.  
 B. Parallel venation is generally found in dicot.  
 C. Reticulate venation is generally found in monocot.  
 D. Reticulate venation is generally found in dicot.

Choose the *most appropriate* answer from the options given below:

- (1) A and D only (2) B and C only  
 (3) B and D only (4) A and C only

129. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R):

**Assertion (A):** Gas vacuoles are found in blue green and purple and green photosynthetic bacteria.

**Reason (R):** Gas vacuoles are not the part of inclusion bodies.

In the light of the above statements, choose the **correct** answer from the options given below.

- (1) A is true but R is false.  
 (2) A is false but R is true.  
 (3) Both A and R are true and R is the correct explanation of A.  
 (4) Both A and R are true, but R is the not correct explanation of A.

130. What are the product of anaerobic respiration in yeast?

- (1) Oxygen, carbon dioxide, water and energy  
 (2) Oxygen, carbon dioxide, ethyl alcohol and energy  
 (3) Carbon dioxide, ethyl alcohol and energy  
 (4) Water, Carbon dioxide, ethyl alcohol and energy

131. Find the **incorrect** Statement-In relation to glycolysis.

- (1) Occur in all organisms.  
 (2) Generates net 2ATP from one glucose.  
 (3) Occur in mitochondria and cytoplasm.  
 (4) Involve 2 substrate level phosphorylation.

132. Consider the following figure and identify the **correct** phase of M phase.



- (1) Metaphase (2) Anaphase  
 (3) Prophase (4) Telophase

133. Which one of the following statements regarding post-fertilization development in flowering plants is **incorrect**?
- (1) Ovary develops into fruit.
  - (2) Zygote develops into embryo.
  - (3) The PEN undergoes successive nuclear divisions.
  - (4) Ovules develop into embryo sac.
134. Introduction of an alien DNA into plants is achieved by:
- (1) competent with bivalent ions.
  - (2) using microinjection.
  - (3) using a gene gun.
  - (4) using lysozymes and chitinase.
135. The relation between species richness and area for a wide variety of taxa (angiosperm plants, birds, bats, freshwater fishes) turns out to be a;
- (1) rectangular parabola.
  - (2) rectangular hyperbola.
  - (3) straight line/linear.
  - (4) sigmoid.

**Botany : Section-B (Q. No. 136 to 150)**

136. Among the following which one confers certain unique phenotypic characters such as resistance to antibiotics in some bacteria?
- (1) Plasmid DNA
  - (2) Ribosomes
  - (3) Mesosomes
  - (4) Genomic DNA
137. Identify the role of filiform apparatus that is found in ovule of flowering plants.
- (1) It guides the entry of pollen tube into synergids.
  - (2) It recognises the suitable pollen grain that falls on the stigma.
  - (3) It produces nectar.
  - (4) It stimulates division of the generative cell.

**138. Match List-I with List-II.**

| List-I             | List-II                 |
|--------------------|-------------------------|
| A. Citric acid     | I. <i>Trichoderma</i>   |
| B. Cyclosporin - A | II. <i>Clostridium</i>  |
| C. Statins         | III. <i>Aspergillus</i> |
| D. Butyric acid    | IV. <i>Monascus</i>     |

Choose the **correct** answer from the options given below:

- (1) A-I, B-IV, C-II, D-III
- (2) A-III, B-IV, C-I, D-II
- (3) A-III, B-I, C-II, D-IV
- (4) A-III, B-I, C-IV, D-II

**139. Agar used in the preparations of ice-creams and jellies are obtained from;**

- (1) *Gelidium* and *Gracilaria*.
- (2) *Gelidium* and *Porphyra*.
- (3) *Porphyra* and *Gracilaria*.
- (4) *Gelidium* and *Sargassum*.

**140. Identify the correct statement regarding geitonogamy.**

- (1) Pollination is achieved within the same flower.
- (2) Geitonogamy is functionally self-pollination.
- (3) It does not involve pollinating agent.
- (4) Genetically it is similar to autogamy since the pollen grains come from the same plant.

**141. Some vascular bundles are described as open because these;**

- (1) are not surrounded by pericycle.
- (2) are surrounded by pericycle but no endodermis.
- (3) are capable of producing secondary xylem and phloem.
- (4) possess conjunctive tissue between xylem and phloem.



142. Given below are two statements:

**Statement-I:** The process of copying genetic information from one strand of the DNA into RNA is termed as transcription.

**Statement-II:** A transcription unit in DNA is defined primarily by the two regions in the DNA. In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement-I is correct but Statement-II is incorrect.
- (2) Statement-I is incorrect but Statement-II is correct.
- (3) Both Statement-I and Statement-II are correct.
- (4) Both Statement-I and Statement-II are incorrect.

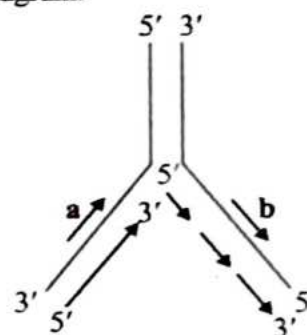
143. Identify the **incorrect** statement(s).

- A. Pollination by wind is more common amongst abiotic pollinations.
- B. Wind-pollination is rare in grasses.
- C. Pollination by water is quite rare in flowering plants and is limited to about 30 genera, mostly monocotyledons.
- D. Only a few flowering plants use a range of animals as pollinating agents.

Choose the *most appropriate* answer from the options given below:

- (1) B and C
- (2) B and D
- (3) A and C
- (4) A and B

144. Select the correct labelling for **a** and **b** from the given diagram.



- | a                           | b                       |
|-----------------------------|-------------------------|
| (1) Continuous synthesis    | Discontinuous synthesis |
| (2) Discontinuous synthesis | Template DNA            |
| (3) Template DNA            | Continuous synthesis    |
| (4) Template DNA            | Discontinuous synthesis |

145. Given below are two statements:

**Statement-I:** Law of segregation is based on the fact that the alleles do not show any blending.

**Statement-II:** A gamete receives only one of the two factors during segregation.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement-I is correct but Statement-II is incorrect.
- (2) Statement-I is incorrect but Statement-II is correct.
- (3) Both Statement-I and Statement-II are correct.
- (4) Both Statement-I and Statement-II are incorrect.

146. Which of the following statement(s) is **correct** w.r.t translation.?

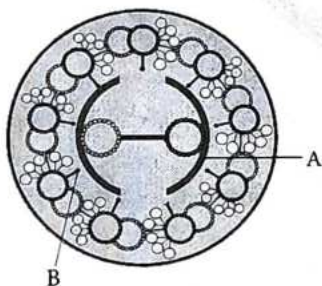
- A. The amino acids are joined by a peptide bond.

- B. The ribosome consists of structural RNAs and about 80 different proteins.
- C. The cellular factory responsible for synthesising proteins is the ribosome.
- D. The presence of a catalyst would enhance the rate of hydrogen bond formation.

Choose the *most appropriate* answer from the options given below.

- (1) A and B only
- (2) All except D
- (3) A, C and D only
- (4) B and C only
147. In which of the following both pairs have **correct** combination?
- (1) *In situ* conservation: Seed Bank  
*Ex situ* conservation: National Park
- (2) *In situ* conservation: Tissue culture  
*Ex situ* conservation: Sacred groves
- (3) *In situ* conservation: National Park  
*Ex situ* conservation: Botanical Garden
- (4) *In situ* conservation: Cryopreservation  
*Ex situ* conservation: Wildlife Sanctuary

148. Identify A and B respectively in the given figure.



- (1) Radial spoke; Hub
- (2) Hub; Radial spoke
- (3) Central sheath; Radial spoke
- (4) Radial spoke; Central sheath

149. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R):

**Assertion (A):** Mutation that arises due to the change in a single base pair of DNA is known as point mutation.

**Reason (R):** Deletions and insertions of one or two bases of DNA, causes frame-shift mutations. In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

150. Match List-I with List-II.

| List-I          | List-II                      |
|-----------------|------------------------------|
| A. Meristem     | I. Photosynthesis, storage   |
| B. Parenchyma   | II. Mechanical support       |
| C. Collenchyma  | III. Actively dividing cells |
| D. Sclerenchyma | IV. Sclereids                |

Choose the **correct** answer from the options given below:

- (1) A-IV, B-II, C-III, D-I
- (2) A-III, B-I, C-II, D-IV
- (3) A-II, B-I, C-III, D-IV
- (4) A-I, B-II, C-III, D-IV

**Zoology : Section-A (Q. No. 151 to 185)**

151. Signals from a fully developed foetus and placenta ultimately lead to parturition which requires the release of:
- (1) estrogen from growing follicles.
- (2) oxytocin from maternal pituitary.
- (3) oxytocin from foetal pituitary.
- (4) relaxin from Leydig cells.



152. Which of the following hormones is used in transplantation surgery to suppress immune response and thus chances of organ rejection by the recipient's body can be avoided?

- (1) Thyroxine
- (2) Cortisol
- (3) Aldosterone
- (4) Steroid hormone

153. Match List-I with List-II:

| List-I                                      | List-II                      |
|---------------------------------------------|------------------------------|
| (A) Origin of the universe                  | (I) 4.5 billion years ago    |
| (B) Origin of earth                         | (II) 3 billion years ago     |
| (C) First non-cellular life form appearance | (III) 2000 million years ago |
| (D) First cellular life form appearance     | (IV) 20 billion years ago    |

Choose the **correct** answer from the options given below:

- (1) A- I, B-IV, C-II, D-III
- (2) A-IV, B-I, C-II, D-III
- (3) A-IV, B-II, C-I, D-III
- (4) A-IV, B-I, C-III, D-II

154. Which of the following sense organs in frogs are composed primarily of cellular aggregations around nerve endings?

- (a) Eyes
- (b) Tympanum with internal ears
- (c) Sensory papillae
- (d) Taste buds
- (e) Nasal epithelium
- (1) (c), (d), and (e) only
- (2) (a), (b), and (c) only
- (3) (d) and (e) only
- (4) (b), (c), and (d) only

155. Which of the following statements is **incorrect**?

- (1) An ideal contraceptive should be user-friendly, easily available and effective.
- (2) Natural methods work on the principle of avoiding chances of ovum and sperm meeting.
- (3) Oral administration of small doses of either progestogens or progestogen-estrogen combinations is another contraceptive method used by females.
- (4) The Medical Termination of Pregnancy (Amendment) Act was enacted in 2019 by the government of India to reduce the incidence of illegal abortion.

156. Given below are two statements:

**Statement-I:** The ventral portion of the midbrain consists mainly of two round swellings called corpora quadrigemina.

**Statement-II:** Midbrain, pons and medulla form the brain stem.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

- (1) Statement-I is correct but Statement-II is incorrect.
- (2) Statement-I is incorrect but Statement-II is correct.
- (3) Both Statement-I and Statement-II are correct.
- (4) Both Statement-I and Statement-II are incorrect.

157. Which of the following is the most accepted definition of biotechnology by the European Federation of Biotechnology?

- (1) Maintenance of sterile ambience for enabling growth of desired microbe/eukaryotic cell in large quantities.

- (2) Technique of using live organisms or enzymes from organisms to produce products and processes useful to animals.
- (3) Process which uses genetically engineered animals only on a large scale for the benefit of mankind.
- (4) The integration of natural science and organisms, cells, parts thereof and molecular analogous for products and services.

**158.** Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R):

**Assertion (A):** All amino acids contain a hydroxyl group and an acidic group as substituents on the same carbon.

**Reason (R):** In amino acids, the  $\alpha$ -carbon has four substituent groups - a hydrogen atom, a carboxyl group, an amino group, and a variable group.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

**159.** Arrange the following events in the order they occur in the Renin-Angiotensin mechanism:

- (a) Conversion of angiotensinogen to angiotensin I and then to angiotensin II
- (b) Release of renin by JG cells
- (c) Activation of adrenal cortex to release Aldosterone
- (d) Increase in blood pressure
- (e) Reabsorption of  $\text{Na}^+$  and water from the distal parts of the tubule

Choose the correct sequence from the options given below:

- (1) (a)-(b)-(d)-(c)-(e)
- (2) (a)-(b)-(c)-(d)-(e)
- (3) (b)-(a)-(c)-(e)-(d)
- (4) (b)-(a)-(c)-(d)-(e)

**160.** Read the following and identify the number of **incorrect** statement(s) related to biological classification of humans.

- A. They belong to the genus *Homo*.
- B. They belong to the class Primata.
- C. They belong to the order Mammalia.
- D. They belong to the phylum Chordata.

- (1) One
- (2) Two
- (3) Three
- (4) Four

**161.** Given below are some statements (A-D).

- A. During swallowing glottis can be covered by a thick elastic cartilaginous flap called epiglottis.
- B. The exchange part is the site of the actual diffusion of  $\text{O}_2$  and  $\text{CO}_2$  between blood and atmospheric air.
- C. The conducting part transports the atmospheric air to the alveoli.
- D. The lungs are situated in the abdominal cavity.

Choose the **incorrect** statements from the options given below:

- (1) A and C only
- (2) B and D only
- (3) C and D only
- (4) A and D only

**162.** Which of the following statements is **correct** regarding DNA polymerase used in PCR?

- (1) It is isolated from a virus.
- (2) It remains active at high temperatures.
- (3) It is used to ligate introduced DNA in recipient cells.
- (4) It serves as a selectable marker.

**163.** Given below are some statements (A-D).

- A. The heart is an endodermally derived organ.
- B. In an adult human, the size of a heart is of clenched fist.



- C. It is protected by a double-walled membranous bag-called the pericardium.  
 D. The small upper two chambers of the human heart are called ventricles.

Choose the **correct** statements from the options given below:

- (1) B and C only (2) A, B and C only  
 (3) A, B and D only (4) A, B, C and D

164. Match List-I with List-II:

| List-I            | List-II             |
|-------------------|---------------------|
| (A) Filarial worm | (I) Cnidaria        |
| (B) Canal system  | (II) Echinodermata  |
| (C) Nematocysts   | (III) Aschelminthes |
| (D) Brittle star  | (IV) Porifera       |

Choose the **correct** answer from the options given below:

- (1) A-IV, B-III, C-II, D-I  
 (2) A-II, B-I, C-IV, D-III  
 (3) A-III, B-IV, C-I, D-II  
 (4) A-II, B-I, C-IV, D-III

165. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R):

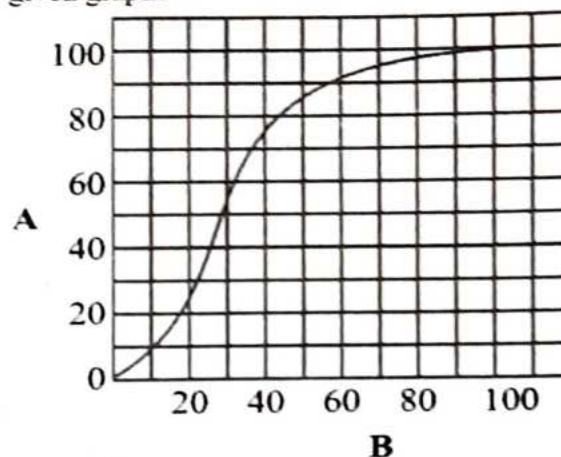
**Assertion (A):** Spermatogenesis starts at the age of puberty due to a significant increase in the secretion of gonadotropin-releasing hormone (GnRH).

**Reason (R):** FSH acts on the Sertoli cells and stimulates the secretion of some factors that help in the process of spermiogenesis.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.  
 (2) A is false but R is true.  
 (3) Both A and R are true and R is the correct explanation of A.  
 (4) Both A and R are true but R is NOT the correct explanation of A.

166. Which of the following is **incorrect** about the given graph?



- (1) The curve is called the oxygen dissociation curve.  
 (2) The part 'A' represents percentage saturation of haemoglobin with oxygen.  
 (3) The part 'B' represents partial pressure of carbon dioxide.  
 (4) This curve is highly useful in studying the effect of factors like  $p\text{CO}_2$ ,  $\text{H}^+$  concentration, etc., on the binding of  $\text{O}_2$  with haemoglobin.

167. An excessive loss of fluid from the body can activate osmoreceptors which stimulate the hypothalamus to release \_\_\_X\_\_\_ from the \_\_\_Y\_\_\_.

Choose the option which fills the blanks **correctly**.

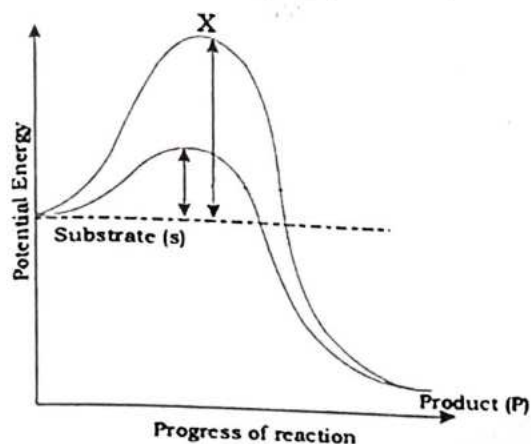
- (1) X- atrial natriuretic factor (ANF);  
 Y- neurohypophysis.  
 (2) X- antidiuretic hormone (ADH);  
 Y- adenohypophysis.  
 (3) X- antidiuretic hormone (ADH);  
 Y- neurohypophysis.  
 (4) X- atrial natriuretic factor (ANF);  
 Y- adenohypophysis.

168. Read the following statements (A-E).
- A. Decline in death rate.
  - B. The rapid increase in MMR and IMR
  - C. Rapid decline in MMR and IMR
  - D. Increase in the number of people in the reproductive age group.
  - E. The rapid increase in the death rate.
- Choose the option that includes factors that help in population growth.
- (1) A, B and D only
  - (2) A, C and D only
  - (3) C, D and E only
  - (4) A, B, C, D and E
169. A competitive inhibitor of succinate dehydrogenase is:
- (1) glutamic acid.
  - (2) malonate.
  - (3) succinate.
  - (4) serine.
170. Multipolar neurons are found in the:
- (1) retina of eye.
  - (2) embryonic stage.
  - (3) cerebral cortex.
  - (4) Both (1) and (2)
171. Place the following events in chronological order according to Darwin's theory of evolution:
- (a) Extinction of those species that are less adapted
  - (b) Appearance of new life forms
  - (c) Variations within a population
  - (d) Natural selection
- Choose the correct sequence from the options given below:
- (1) (c), (d), (b), (a)
  - (2) (a), (b), (c), (d)
  - (3) (d), (c), (a), (b)
  - (4) (b), (a), (d), (c)

172. Given below are two statements:
- Statement-I:** Muscle contraction is initiated by a signal sent by the central nervous system via a sensory neuron.
- Statement-II:** Scapula is a large triangular flat bone situated in the dorsal part of the thorax between the second and the seventh ribs.
- In the light of the above statements, choose the *most appropriate* answer from the options given below:
- (1) Statement-I is correct but Statement-II is incorrect.
  - (2) Statement-I is incorrect but Statement-II is correct.
  - (3) Both Statement-I and Statement-II are correct.
  - (4) Both Statement-I and Statement-II are incorrect.
173. Which of the following statements is/are **correct** regarding the process by which PTH increases  $\text{Ca}^{2+}$  level in blood?
- (1) By acting on bones and stimulating the process of bone resorption.
  - (2) By reabsorption of  $\text{Ca}^{2+}$  from the renal tubules.
  - (3) By increasing  $\text{Ca}^{2+}$  absorption from the digested food.
  - (4) All of these
174. Which of the following characteristics are unique to the kingdom Animalia?
- (a) Multicellular
  - (b) Heterotrophic
  - (c) Cell walls
  - (d) Holozoic nutrition
  - (e) Production of spores
  - (f) Copulation
- (1) (a), (b) and (c) only
  - (2) (a), (b), (d) and (f) only
  - (3) (b), (c), (d) and (e) only
  - (4) (c), (d) and (f) only



175. Which of the following statements is **correct** for 'X' in the given diagram?



- (1) It is the transition state.  
 (2) It is a competitive inhibitor.  
 (3) It is a cofactor.  
 (4) It is an apoenzyme.
176. Red muscle fibres:
- (1) contain less quantity of myoglobin.  
 (2) possess a high amount of sarcoplasmic reticulum.  
 (3) only depend on anaerobic processes for energy.  
 (4) can utilise a large amount of oxygen for ATP production due to plenty of mitochondria.
177. Which of the following statements are correct about thermoregulation?
- (a) Thermoregulation is energetically expensive for many large animals.  
 (b) Small animals have a larger surface area relative to their volume.  
 (c) Small animals tend to lose body heat very fast when it is cold outside.  
 (d) For smaller organisms maintaining body temperature requires less energy.

Select the correct option:

- (1) (a), (b), (c)      (2) (b), (c)  
 (3) (a), (b)          (4) (a), (c), (d)

178. Given below are two statements:

**Statement-I:** Lymph is a colourless fluid containing specialised erythrocytes which are responsible for the immune system of the body.

**Statement-II:** Plasma transports  $\text{CO}_2$  and RBCs transport oxygen only.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement-I is correct but Statement-II is incorrect.  
 (2) Statement-I is incorrect but Statement-II is correct.  
 (3) Both Statement-I and Statement-II are correct.  
 (4) Both Statement-I and Statement-II are incorrect.

179. In which of the following phylums' segmentation in the body is observed first?

- (1) Aschelminthes      (2) Arthropoda  
 (3) Annelida            (4) Platyhelminthes

180. Which one of the following is an **incorrect** match?

- (1) Myometrium - Exhibits strong contractions during delivery of the baby  
 (2) Endometrium - Undergoes cyclical changes during menstrual cycle  
 (3) Perimetrium - External thin membranous of uterus  
 (4) Uterus - Birth canal

181. Match **List-I** with **List-II**:

| List-I                    | List-II                                                    |
|---------------------------|------------------------------------------------------------|
| (A) Mushroom gland        | (I) 6 <sup>th</sup> to 8 <sup>th</sup> abdominal segment   |
| (B) Ovaries               | (II) 6 to 8 in number                                      |
| (C) A pair of spermatheca | (III) 2 <sup>nd</sup> to 6 <sup>th</sup> abdominal segment |
| (D) Hepatic caeca         | (IV) 6 <sup>th</sup> segment                               |

Choose the **correct** answer from the options given below:

- (1) A- IV, B-II, C-III, D-I
- (2) A-III, B-II, C-IV, D-I
- (3) A-II, B-III, C-I, D- IV
- (4) A-I, B-III, C-IV, D- II

**182.** Human immunodeficiency virus (HIV) has a protein coat and a genetic material, that is:

- (1) one double-stranded RNA.
- (2) two double-stranded DNA.
- (3) one single-stranded DNA.
- (4) two single-stranded RNA.

**183.** Which one of the following statements is **incorrect** about the menstrual cycle?

- (1) The first menstruation begins at puberty and is called menarche.
- (2) Lack of menstruation may also occur due to some environmental factors like stress, and poor health.
- (3) The corpus luteum secretes large amounts of progesterone which is essential for the maintenance of endometrium.
- (4) In the presence of fertilisation, the corpus luteum degenerates in the luteal phase.

**184.** Consider the following effects related to ADH activity.

Which combination correctly lists the effects when there is an increase in ADH due to a decrease in blood volume?

- (a) Constriction of blood vessels
- (b) Increase in urine output
- (c) Decrease in blood pressure
- (d) Increase in water reabsorption from the kidneys
- (e) Increase in glomerular blood flow

Choose the correct answer from the options given below:

- (1) (a), (d), and (e) only
- (2) (a), (b), and (c) only
- (3) (a), (d), and (c) only
- (4) (b), (c), and (e) only

**185.** Which of the following are related to protein structure?

- (a) Glycosidic bond
- (b) Homopolymer
- (c) Disulphide bond
- (d) Hydrogen bond
- (e) Phosphodiester bond
- (f) Lecithin
- (g) Haemoglobin
- (1) (a), (b), (c), (d)
- (2) (c), (d), (g)
- (3) (d), (e), (f), (g)
- (4) (b), (d), (e), (g)

**Zoology : Section-B (Q. No. 186 to 200)**

**186.** Given below are two statements:

**Statement-I:** The first amphibians on land and water were lobefins.

**Statement-II:** Lobefins were ancestors of salamanders.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement-I is correct but Statement-II is incorrect.
- (2) Statement-I is incorrect but Statement-II is correct.
- (3) Both Statement-I and Statement-II are correct.
- (4) Both Statement-I and Statement-II are incorrect.

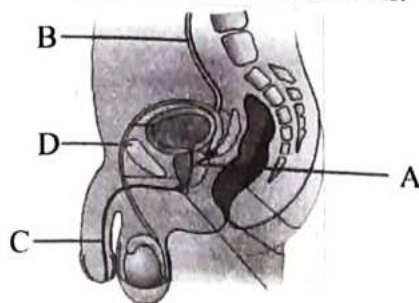
**187.** Globulins present in human blood plasma are primarily involved in:

- (1) osmotic balance of body fluids.
- (2) oxygen transport in the blood.
- (3) clotting of blood.
- (4) defense mechanisms of body.





188. It is a diagrammatic sectional view of the male reproductive system. Identify the ejaculatory duct which is formed by the fusion of a duct of seminal vesicle and vas deferens.



- (1) A (2) B  
(3) D (4) C
189. Which of the following statements is **correct** about repolarisation?
- (1) Sodium ions rapidly move inside the cell.
  - (2) Sodium ions rapidly move out of the cell.
  - (3) Potassium ions rapidly move inside the cell.
  - (4) Potassium ions rapidly move out of the cell.
190. An American company prepared two DNA sequences corresponding to A and B chains of human insulin and introduced them through a plasmid into:
- (1) *Bacillus thuringiensis*.
  - (2) *Escherichia coli*.
  - (3) *Salmonella typhimurium*.
  - (4) *Haemophilus influenzae*.

191. Match List-I with List-II:

| List-I         | List-II            |
|----------------|--------------------|
| (A) Earthworm  | (I) Moist cuticle  |
| (B) Arthropods | (II) Gills         |
| (C) Fishes     | (III) Lungs        |
| (D) Birds      | (IV) Tracheal tube |

Choose the **correct** answer from the options given below:

- (1) A-II, B-I, C-IV, D-III
- (2) A-I, B-IV, C-II, D-III
- (3) A-I, B-III, C-II, D-IV
- (4) A-I, B-II, C-IV, D-III

192.   A   is the most abundant protein in animal world and   B   is the most abundant protein in the whole biosphere.

Choose the **correct** option to fill in the blanks.

- (1) A – Collagen, B – RuBisCO
- (2) A – Collagen, B – keratin
- (3) A – Keratin, B – RuBisCO
- (4) A – Keratin, B – collagen

193. In a resting neuron, the axonal membrane is:

- (1) nearly impermeable to potassium ions.
- (2) impermeable to positively charged proteins present in the axoplasm.
- (3) comparatively more permeable to  $K^+$ .
- (4) completely permeable to sodium ions.

194. Choose the **correct** pair.

- (1) Lysozyme - Bacterial cell wall digesting enzyme
- (2) Polymerases - Break the DNA into fragments
- (3) Nucleases - Separate the two strands of DNA
- (4) Exonucleases - Make cuts at specific positions within DNA

195. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R):

**Assertion (A):** The alimentary canal of a frog is short compared to a tadpole.

**Reason (R):** This is because frogs are carnivores and tadpoles are herbivorous.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

196. Lippes loop is a:

- (1) diaphragm.
- (2) contraceptive pill.
- (3) non-medicated IUDs.
- (4) hormone-releasing IUDs.

197. Given below are two statements:

**Statement-I:** The osmolarity in the cortex is  $300\text{mOsmolL}^{-1}$  and in the inner medulla is  $1200\text{mOsmolL}^{-1}$ .

**Statement-II:** NaCl is transported by the ascending limb of Henle's loop which is exchanged with the descending limb of *vasa recta*.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement-I is correct but Statement-II is incorrect.
- (2) Statement-I is incorrect but Statement-II is correct.
- (3) Both Statement-I and Statement-II are correct.
- (4) Both Statement-I and Statement-II are incorrect.

198. Which of the following results from hypothyroidism?

- (1) Cretinism
- (2) Addison's disease
- (3) Grave's disease
- (4) Dwarfism

199. Which of the following diseases is an autoimmune disorder?

- (1) Rheumatoid arthritis
- (2) Arthritis
- (3) Osteoporosis
- (4) Gout

200. Given below are some statements (A-E).

- A. When any plane passing through the central axis of the body divides the organism into two identical halves, it is called radial symmetry.
- B. Coelenterates are triploblastic animals.
- C. Animals belonging to the Aschelminthes phylum are pseudocoelomates.
- D. Notochord is a mesodermally derived.
- E. The property of a living organism to emit light is well-marked in ctenophores.

Choose the **correct** statements from the options given below.

- (1) A, B and D only
- (2) A, C and D only
- (3) A, C, D and E only
- (4) B and D only